Provenance

Data Architecture

5-Axis Model · Identity System · Graph Navigation

Transfer Service · Event-Driven Lifecycle

Marine Aquaculture Platform

HOW WE GOT HERE

From Spreadsheets to a Unified Platform

Babelfish

2013–2019 *The first database*

Rails + React + PostgreSQL
Heroku · Single-org

- Basic tank & site tracking
- QR code coral identification
- Mobile camera for observations
- Single organization only
- Coral-specific data model
- No offline support

Hard-coded to coral. One org at a time. No genetics tracking.

→

NewFish

2019–2023 *The candy exercise*

React TypeScript + Docker
Heroku · Improved schema

- Better group/observation models
- AWS S3 image uploads
- Containerized deployment
- Improved breadcrumbs & UI
- Still single-organism (coral)
- No cross-org transfers

Still coral-only. Same fundamental constraints. No multi-org.

→

Provenance

2023–Present *Complete rewrite*

Flutter + Firebase
Mobile-first · Offline-capable

- ✓ 9 organism kinds supported
- ✓ 5-axis composable data model
- ✓ Multi-org with transfers
- ✓ Provenance ID crosswalk
- ✓ Offline-first mobile app
- ✓ Event-driven audit trail

Organism-agnostic. Designed for the entire restoration community.

The candy exercise proved the old model couldn't flex. Provenance was built from scratch to handle any organism, any workflow, any organization.

THE 5-AXIS INVENTORY MODEL

Five Composable Dimensions

Every organism record is expressed through five neutral axes — supporting coral, oyster, kelp, seagrass, and beyond.

1
Organism + Species

What & taxonomy

2
Provenance Type

Origin story

3
Life Stage

Biological phase

4
Physical Form

Handling unit

5
Quantity + Size

Count & measure

Immutable
Axes 1 & 2 set at creation

Mutable + Tracked
Axes 3–5 evolve via events

Organism-Agnostic
Same model across all species

WHY THIS MATTERS

What a Rich Inventory Model Enables

Tracking 5 axes per record — not just species and location — unlocks powerful analytical capabilities across the restoration pipeline.

Growth Rate Tracking

Size bands + life stage transitions over time reveal per-genet growth rates. Compare performance across nursery conditions, sites, and seasons.

Maturation Rate Analysis

Life stage timestamps (juvenile → adult) per organism kind let you measure maturation timelines and identify fast-growing genets.

Genetic Diversity at Outplanting

Because every record carries a genetId, you can measure how many unique genets reached each reef site — critical for resilience.

Volume & Tissue Area Deployed

Physical form + size band data lets you calculate total tissue area and volume outplanted per site, genet, or species.

Provenance-Aware Reporting

Immutable provenance types mean you can report on wild-collected vs. sexually propagated vs. fragmented stock separately.

Cross-Org Comparisons

Standardized axes + the PID crosswalk enable apples-to-apples comparisons of restoration outcomes across organizations.

Without these axes, you're limited to counting organisms. With them, you can measure restoration effectiveness.

FROM DATA TO IMPACT

From Field Work to Funder Reports

The same data powering daily operations also produces the reports funders and regulators need — no separate data entry required.

Daily Operations

Field teams do their work

- Log outplanting events
- Record size measurements
- Track mortality & fragmentation
- Transfer organisms between orgs
- Update life stages & locations

→

Automated Reporting

Data becomes insight

- Survival rates by genet & site
- Growth curves over time
- Genetic diversity per reef
- Total tissue area deployed
- Chain-of-custody audit trails

→

Stakeholder Impact

Demonstrate restoration success

- NFWF grant milestone reporting
- NOAA compliance documentation
- Donor & sponsor updates
- Peer-reviewed data exports
- Community engagement metrics

Practitioners shouldn't have to choose between doing restoration and documenting it. Provenance makes reporting a byproduct of operations.

IDENTITY ARCHITECTURE

Genets vs. Organism Records

GENET — Genetic Lineage

The shared DNA identity across all physical instances.

Field	Description	Scope	Example
localId	Your team's name	Org	ACER-001
provenanceId	Cross-org ID (PID)	Global	PID-AP-0042
id (UUID)	Database primary key	Global	c9a1f2e8...

RECORD — Physical Instance

One fragment, colony, or batch tracked individually.

Field	Description	Scope	Example
recordName	Friendly nickname	Instance	"Crimson"
localId	Genet ID inherited	Inherited	ACER-001
genetId	Link to Genet record	FK	c9a1f2e8...
id (UUID)	This instance's key	Global	d4e7a1b3...

localId + provenanceId = "What genetic lineage?"

recordName + recordUUID = "Which physical organism?"

ONE GENET, MANY RECORDS

A Concrete Example

GENET

ACER-001

Species: *A. cervicornis*

PID: **PID-AP-0042**

Provenance: Wild collection

Cross date: 2023-06-15

genetId

"Crimson" localId: ACER-001

Tank A · Nursery · 15 fragments · adult

"Blaze" localId: ACER-001

Tank B · Nursery · 8 fragments · adult

"Reef Site 12" localId: ACER-001

Outplant Zone C · 1 colony · adult

All three records share the same localId (ACER-001) and genetId — proving they are genetically identical.

Each has its own recordName and recordUUID to distinguish the physical instance.

SEXUAL PROPAGATION & MATURATION

From Gametes to Tracked Individuals

1 Gamete Collection

Broodstock spawns. Eggs + sperm collected in containers.

container

`gamete`

2 Cross Event

Dam × Sire gametes combined. Creates a new cohort.

—

`embryo`

3 Settlement & Growth

Larvae settle on substrates. Counted by density per unit.

shared substrate

`recruit → juvenile`

4 Cohort Management

Batch tracked as a collective. Individuals not yet distinguished.

shared substrate

`juvenile`

5 Graduation Event

Select individuals promoted to solo tracking as adults.

individual

`adult`

Provenance Type changes from cohort → sexual recruit at graduation. genetId remains the same.

Applies to all sexually reproducing organisms: coral, oyster, urchin, crab, finfish, sea cucumber.

`Physical form transitions: container (gametes) → shared substrate (settled cohort) → individual (graduated adult)`

THE 5-AXIS MODEL IN PRACTICE

Real-World Record Examples

Every combination of axes produces a distinct record type — the same model handles all restoration use cases.

Traditional Plug

A graduated coral fragment mounted on a cement plug, tracked individually by name.

Organism / Species	Coral · Acropora cervicornis
localId	ACER-001
recordName	Fluffy
Provenance Type	sexualRecruit
Life Stage	adult
Physical Form	individual · plug
Size Band	M (5–15 cm^2)
Surface Area	12.4 cm^2
Location	Nursery Ex-Situ · Tank 3 · Tree 5

Wild-Collected Clipping

Branch cut from a wild colony during a field collection event. Unmounted fragment.

Organism / Species	Coral · Orbicella faveolata
localId	OFAV-WC-042
recordName	Boulder
Provenance Type	founder
Life Stage	adult
Physical Form	individual · branch
Size Band	L (15–50 cm^2)
Surface Area	38.1 cm^2
Location	Nursery In-Situ · Reef Line A · Tree 12

Cryopreserved Sperm

Frozen sperm in a labeled vial. Size = density per volume. Volumetric counting mode.

Organism / Species	Coral · Acropora palmata
localId	APAL-BS-007
recordName	Straw 7-A
Provenance Type	founder
Life Stage	gamete
Physical Form	container · vial
Size Band	XS
Count	~500 sperm/mL · 0.5 mL
Location	Gene Bank · Cryo Dewar 2

Each card is one OrganismRecord in the database — localId links it to a genet, recordName distinguishes the physical instance.

THE 5-AXIS MODEL IN PRACTICE

More Examples Across Organisms

The 5-axis model is organism-agnostic — oysters, crabs, and other species use the exact same data structure.

Settled Cohort on Tile

Sexually-crossed recruits settled on a ceramic tile. Batch-tracked by density per substrate.

Organism / Species	Coral · Acropora cervicornis
localId	ACER-C24-001
recordName	Tile 14-B
Provenance Type	cohort
Life Stage	recruit → juvenile
Physical Form	sharedSubstrate · tile
Size Band	M
Density	3.2 recruits / tile
Location	Nursery Ex-Situ · Tank 1 · Tray A

Oyster Spat Bag

Batch of juvenile oysters in a mesh bag. High-density batch tracking by container.

Organism / Species	Oyster · Crassostrea virginica
localId	CVIR-H24-003
recordName	Bag 3-East
Provenance Type	cohort
Life Stage	spat
Physical Form	sharedSubstrate · bag
Size Band	L
Count	~3,500 per bag
Location	Reef Aquaculture · Oyster Lease B · Cage 4

Crab Broodstock

Spawning adult crab held in a raceway. Tracked as individual broodstock for breeding program.

Organism / Species	Crab · Callinectes sapidus
localId	CSAP-BS-011
recordName	Blue Queen
Provenance Type	founder
Life Stage	broodstock
Physical Form	individual
Size Band	XL
Carapace Width	18.2 cm
Location	Grow-Out Pond · Raceway 2

The model catches invalid combinations: broodstock cannot be in a container, cohort requires shared substrate or container form.

PROPAGATION, SPLITS & MERGES

How One Genet Becomes Many Records

Every operation creates new records that inherit the same genetId and localId — the genetic thread is never broken.

`ACER-001 · genetId: c9a1f2e8...`

Fragment (Split)

Colony broken into fragments for nursery propagation.

BEFORE

1 colony · 450 cm²

↓

AFTER

- remaining · 200 cm²
- frag-F1 · 80 cm²
- frag-F2 · 80 cm²

3 records

Merge

Fragments from same genet combined back into one tracking unit.

BEFORE

2 records · same genet

↓

AFTER

- merged · 350 cm²

2 → 1 record

Outplant

Fragments distributed to different reef sites, each tracked solo.

BEFORE

10 frags in Tank A

↓

AFTER

- Site Alpha · planted
- Site Beta · planted
- Tank A · remaining

3+ records

Cohort Split

Settled cohort divided across multiple substrates or structures.

BEFORE

1 tile · 50 recruits

↓

AFTER

- Tile-A · 25 recruits
- Tile-B · 25 recruits

2 batch records

All operations preserve the genetId + localId link — the genetic thread is never broken regardless of physical form changes.

PROVENANCE TYPES

How Genetic Relationships Form

Wild

Collected from nature. Becomes broodstock — the foundation.

broodstock

collection method + date

Sexual Cohort

Born from a cross. Links to dam & sire gametes.

juvenile

parent gamete IDs + cross date

Graduated

Promoted from a batch to individual tracking.

adult

source cohort linkage

Transfer

Imported from another organization.

adult

source organization ID

Unknown

Origin not yet documented.

adult

none

Genetic Relationship Types

Parent → Offspring

`damGameteIds + sireGameteIds`

Siblings

`Shared parentGameteIds`

Cohort → Individual

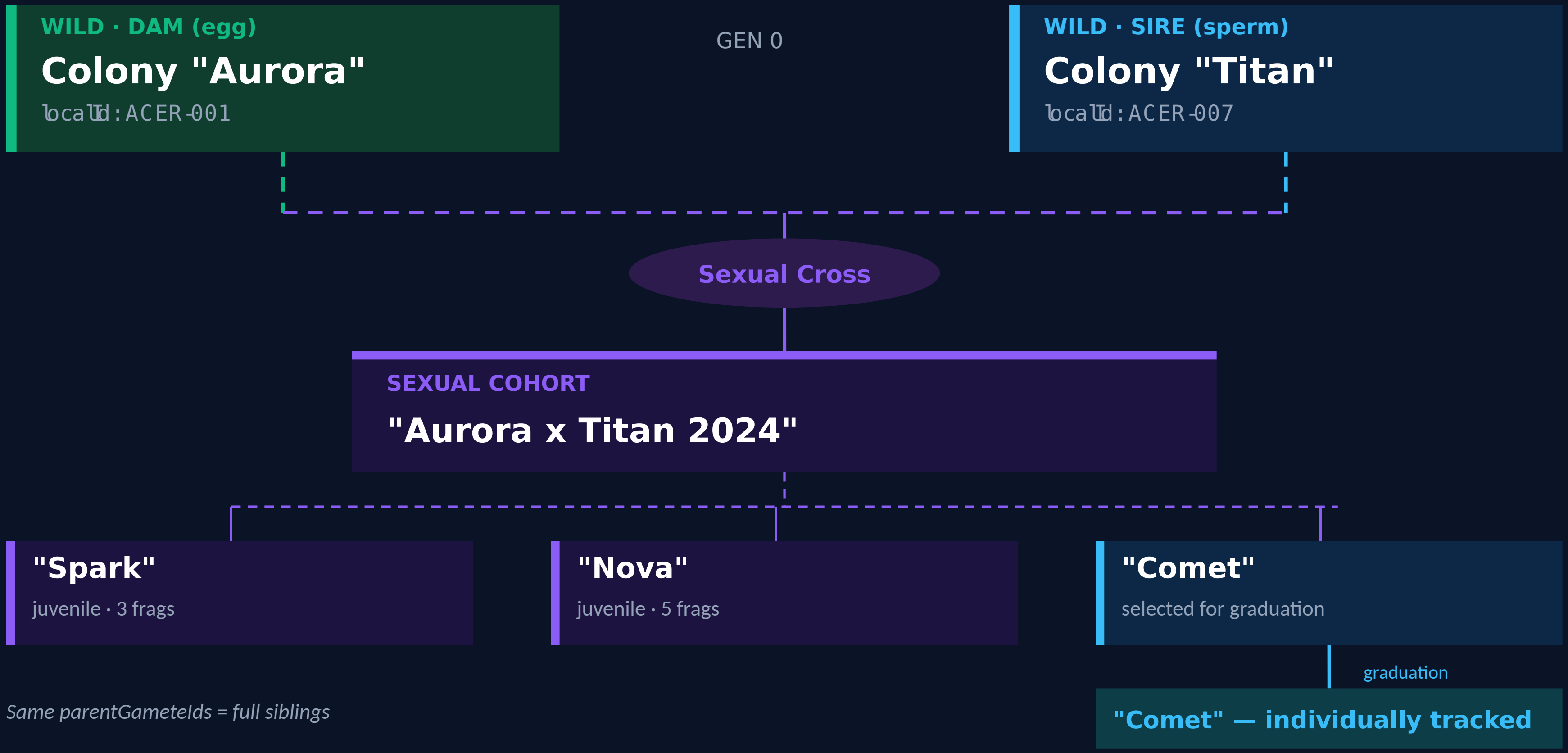
`parentCohortId link`

Clonal (Asexual)

`Same genetId + donorGenotypeId`

PROVENANCE IN ACTION

A Complete Family Tree



GRAPH NODE ARCHITECTURE

Hierarchical Navigation & Reactive Streams

Navigation Hierarchy

OrganizationNode
Root · children: sites

SiteNode
Nursery/reef · children: groups + organisms

GroupNode
Tank/table · MovableNode · nestable

OrganismNode
Record instance · MovableNode

Load State Machine

Initial → Loading → Loaded → Error

4-Stream Bundle (per node)

Record Stream
Real-time record updates from Firestore

Children Stream
Child nodes (sites, groups, organisms)

Events Stream
All descendant events by urlPath prefix

Creator Stream
User who created this record

`CombineLatestStream → LoadedState`

Uri-based navigation internally · urlPath for deep links & queries · Lazy loading: only immediate descendants

SITES & STRUCTURE HIERARCHY

Where Organisms Live

Sites contain a 3-level structure hierarchy. Pro organizations can customize structure types per site.

3-Level Structure Hierarchy

Site
Nursery, reef, lab, gene bank, kelp farm...
"Tavernier Nursery"

Superstructure
Zone or life-support system grouping
"Zone A — Broodstock"

Structure
Tank, tree, table, raceway, dome, cage...
"Tank 1" · "Tree 5"

Substructure
Tray, branch, grid cell, tag, bag, rack...
"Tray 1-A" · "Branch 3"

Organisms attach to structures and substructures only

Site Types (15+)

Nursery (ex situ) Land-based tanks, raceways	Land
Nursery (in situ) Ocean trees, tables, cradles	Ocean
Outplanting Natural reef restoration areas	Ocean
Gene Bank Genetic preservation facility	Land
Kelp Farm Longline aquaculture in open ocean	Ocean
Reef Aquaculture Oyster/shellfish reef farming	Ocean
Seagrass Plot Restoration and monitoring plots	Ocean
Grow-Out Pond Enclosed pond for crab, urchin, finfish	Land
Raceway Flow-through finfish culture	Land
Release Site Wild release location tracking	Ocean

Pro organizations can customize allowed structure types per site type and add organism-specific structures (domes, cages, longlines, bags).

EVENT-DRIVEN ARCHITECTURE

How Events Modify Records

1 User Action

Change life stage,
move location,
add quantity,
update size

→

2 Detect Changes

ChangeService
compares before/
after field diffs

→

3 Emit Event

Immutable event
stored in Firestore
with snapshotData

→

4 Update Record

Record updated,
history appended,
stream emits

Event Categories (40+ types)

Lifecycle

LifeStageTransition
PhysicalForm Change
SizeChange
QuantityChange

Inventory

Create · Move
Mortality
Collection
Split · Merge

Observation

Disease · Bleaching
Biofouling
Maintenance
Monitoring

Specialized

Husbandry
Reproductive
Environmental
Transfer

Every event stores snapshotData = complete record state at that moment → enabling "time travel" to any point in history

INTER-ORG TRANSFER SERVICE

Cross-Organization Transfers

Transfer Status State Machine

Draft → Pending → Shipped → Received

Rejected | Cancelled

(terminal states — any non-terminal can reach these)

Three Ownership Models

Full Transfer

Receiver becomes owner + manager.
Complete handoff of all rights.

Retained Ownership

Sender keeps ownership.
Receiver manages day-to-day.

Third-Party Transfer

Auto-detected when organism's owner differs from sender.

Transfer Manifest

Point-in-time snapshot for verification & cross-org sharing

- From/To Org Snapshots
- Genet Snapshot
- SHA256 Checksum
- QR Payload
- Status History
- Custody Chain

CHAIN OF CUSTODY

Tracking Ownership Across Organizations

2023-06 → 2024-03

Original creation

Owner: **Reef Lab**

Manager: **Reef Lab**

Wild collection by Reef Lab

2024-03 → 2025-01

retainedOwnership

Owner: **Reef Lab**

Manager: **Ocean Nursery**

Reef Lab retains ownership,
Ocean Nursery manages daily care

2025-01 → present

fullTransfer

Owner: **Ocean Nursery**

Manager: **Ocean Nursery**

Full ownership transferred
to Ocean Nursery

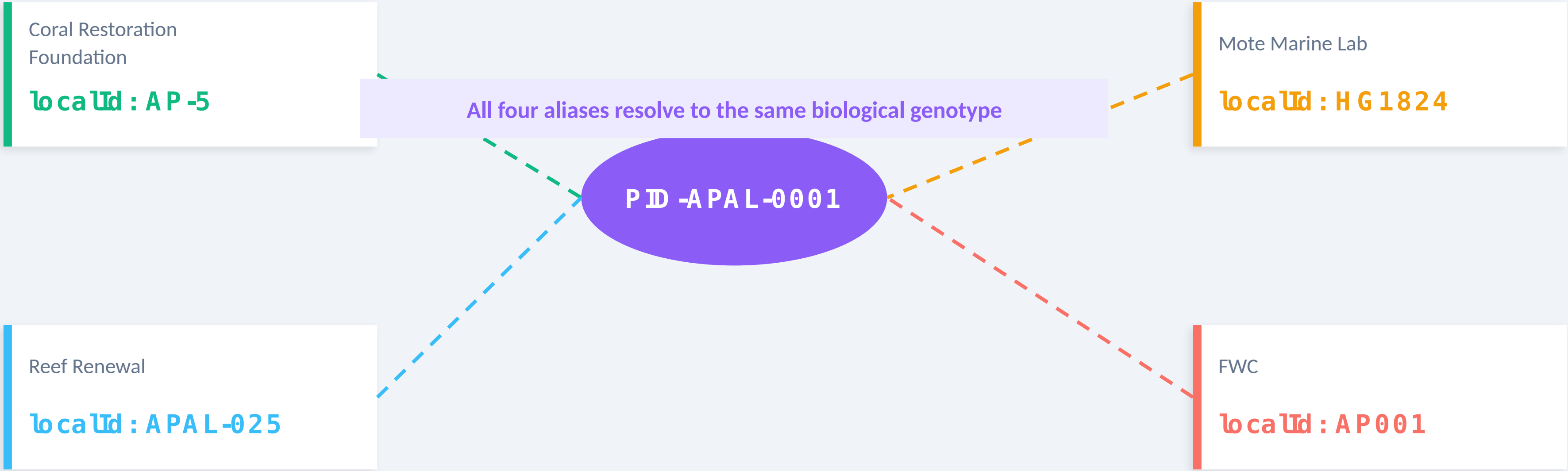
CustodyHistoryEntry Fields

`ownerOrgId`	Collection owner	`managingOrgId`	Day-to-day manager	`startedAt`	Period start date
`transferId`	Triggering transfer	`transferType`	Ownership model	`note`	Description text

PROVENANCE ID CROSSWALK

One Genotype, Many Names, One PID

Every organization uses its own local naming convention — the crosswalk service maps them all to a single Provenance ID.



LocalId Rename — PID Preserved

CRF renames AP-5 → APAL-CRF-5 → alias_index updated atomically → PID-APAL-0001 unchanged

RECORD METADATA

What Lives on an Organism Record

Identity

recordName
localId · genetId
speciesId · organismKind

Location

siteId · groupId
zoneId · subplotId
tagId

Lifecycle State

lifeStage (LifeStageSpec)
physicalForm
sizeSpec · measurement

Provenance

provenanceType
provenanceAttributes
(dam/sire/cohort refs)

Ownership

ownerOrganizationId
managingOrganizationId
custodyHistory[]

References

foreignKeys map
aliases list
completeness score

Completeness Score (0–100%): Tracks populated fields per record, surfacing data gaps in the UI

NFWF-FUNDED DEVELOPMENT

Open Source Foundation for Restoration

NFWF investment builds the freely available core that every restoration practitioner can use.

COMMUNITY TIER — Free, Open Source, Web-Based

Multi-Species Inventory

5-axis model supporting coral, oyster, kelp, seagrass — any marine organism

Provenance Tracking

Complete genetic lineage: wild collection, sexual crosses, cohorts, graduation

Outplanting Management

Track organisms from nursery to reef with site/zone mapping

Basic Monitoring

Growth, survival, health observations at outplant sites

Inter-Org Transfers

Share organisms across organizations with full chain of custody

Identity Architecture

Genet + Record dual identity system for precise genetic tracking

NFWF funding creates shared infrastructure that lowers the barrier for every restoration program.

No license fees. No vendor lock-in. Interoperable by design.

DEVELOPMENT STRATEGY

Open Source + Commercial Services

NFWF-funded community features form the base. Commercial tiers fund ongoing development and hosted services.

	Community NFWF Funded · Free	Pro Commercial · $	Scale Enterprise · $$
Inventory (5-axis model)	✓	✓	✓
Outplanting + Basic Monitoring	✓	✓	✓
Inter-Org Transfers	✓	✓	✓
Provenance ID System	✓	✓	✓
Mobile App + Offline Sync	—	✓	✓
Full Monitoring (KML/Shapefiles)	—	✓	✓
Husbandry + Observations	—	✓	✓
Image Hosting + Comments	—	✓	✓
Task Management + Scheduling	—	—	✓
Sebastian AI Assistant	—	—	✓
Workforce + Training Modules	—	—	✓
Permit/Grant Reporting	—	—	✓

Commercial revenue funds continued open-source development, infrastructure maintenance, and community support.

CENTRALLY HOSTED SERVICES

Shared Infrastructure for the Community

Some capabilities require centrally maintained services that benefit all tiers.

Provenance ID Service

All Tiers

Global PID registry ensuring unique, cross-organization genetic identifiers. Every PID (PID-XX-NNNN) is minted centrally to prevent collisions across the entire restoration community.

Community Feed

All Tiers

Shared knowledge board connecting restoration practitioners. Post updates, share outplanting results, coordinate across organizations — a professional network for marine restoration.

Firebase Cloud Infrastructure

All Tiers

Authentication, real-time database sync, cloud functions for transfer orchestration, and storage for images and documents. Foundation for offline-first architecture.

Image Hosting + Storage

Pro+

Centralized media storage for organism photos, monitoring images, and documentation. Enables visual tracking of health, growth, and outplant success over time.

Sebastian AI

Scale

AI-powered assistant trained on restoration best practices. Helps with data entry, anomaly detection, and operational recommendations for Scale-tier organizations.

Permit/Grant Reporting

Scale

Automated report generation for NFWF deliverables, ESA permits, and grant compliance. Pulls real inventory and monitoring data into structured templates.

Hosted services are sustained through commercial tier subscriptions, ensuring long-term viability without grant dependency.

THE BIGGER PICTURE

NFWF Investment as a Catalyst

Phase 1 — NFWF Foundation

Current

- Core inventory + provenance system
- Inter-org transfer service
- Basic monitoring + outplanting
- Open-source web platform

→

Phase 2 — Practitioner Adoption

Next

- Mobile app for field work
- Offline-first data collection
- Advanced monitoring + husbandry
- Image hosting + collaboration

→

Phase 3 — Ecosystem Scale

Future

- AI-assisted operations
- Automated compliance reporting
- Workforce management + training
- Cross-platform analytics

Why This Matters for NFWF

- Grant investment creates permanent public infrastructure, not throwaway tools
- Commercial tiers fund maintenance of the free community platform
- Standardized data model enables cross-program analytics for NFWF

Provenance

Open Source Core · Hosted Services · Commercial Sustainability

5-Axis Model · Dual Identity · Graph Navigation

Event-Driven Lifecycle · Cross-Org Transfers

NFWF-funded infrastructure enabling restoration at scale.